Polynomial calculus degree above p for the onto functional pigeonhole principle with surplus divisible by p squared

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1 Introduction

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Setting

The onto functional pigeonhole principle onto-FPHP$_n^{n+k}$ says that $n+k$ pigeons can be mapped one-to-one onto n holes; it is false for every $k \geq 1$. In Boolean variables x_{ij} (pigeon i in hole j) its axioms are $\sum_j x_{ij} - 1$, $\sum_i x_{ij} - 1$, $x_{ij}x_{i'j}$ and $x_{ij}x_{ij'}$, the linear encoding used for this family [1, 2]. These results are for the linear encoding only: translating the CNF clause by clause instead turns each pigeon clause into a polynomial of degree n [2], so degree bounds below n say nothing about that encoding. A polynomial calculus (PC) refutation over $\mathbb{F}_p$ derives 1 from the axioms and the Boolean axioms $x^2 - x$ by linear combinations and multiplication by a variable; its degree is the largest degree of a line.

When p does not divide k there is a degree-1 refutation. When p divides k but p^2 does not, the PC degree is exactly p for $n \geq 2p-3$ (a companion finding on this site). This finding is about surplus divisible by p^2.

Result

For every prime p, every $c \geq 1$, and every $n \geq 2p-1$, onto-FPHP$_n^{n+cp^2}$ has no PC refutation over $\mathbb{F}_p$ of degree at most p. Its PC degree is therefore at least $p+1$. With the companion result, surplus divisible by p^2 is strictly harder for PC over $\mathbb{F}_p$ than surplus divisible only by p, for every prime and from a linear threshold in n.

The proof constructs, for every prime p, an explicit integer matrix on $2p-1$ rows and columns (a "square lift"). From it comes a balanced design, which serves as a degree-p witness that no refutation of that degree exists. The cases

$p = 2$ ($n \geq 3$, surplus $4c$) and $p = 3$ ($n \geq 5$, surplus $9c$) are also proved separately.

Why it matters

Razborov's Question 7 asks whether PC degree for surplus p^ℓ stays bounded in characteristic p [2]. At $\ell = 2$ the known PC lower bound is 3, for $N \geq 2(21 + 3p^2) - p$ [4]. This result gives $p + 1$ from $n = 2p - 1$: larger for every $p \geq 3$, and at a far smaller n. Earlier runs of this programme proved $p + 1$ for $p = 3, 5, 7$ separately; this result covers every prime at once.

Proved vs checked

Everything is proved in Lean 4 and kernel-checked, for all primes p, all c and all $n \geq 2p - 1$; nothing is checked only up to a bound. The proofs use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The main theorem is stated for standard (syntactic) PC with Boolean axioms. A second agent in the run, working separately from the author, re-proved all six statements in Lean. No person has reviewed the proofs; the evidence is the kernel check and the independent re-proof.

Novelty statement

This result extends:

- Riis-Sitharam, ECCC TR97-048, Theorem 33: PC degree >= 2^l - 1 for onto-PHP^{N+p^l}_N over characteristic p once N >= (21+3p^2)(2^l - 2) - p (same family; proof not checked by us; at l = 2 this is >= 3 against our >= p+1 from n = 2p-1; at p = 2, l = 3 it is >= 7 for N >= 196 against our >= 5 for n >= 7).
- Beame-Riis 1998, Theorem 8.1 and Lemma 8.7: Nullstellensatz degree between 2^l - 1 and p^l - 1 (multiplier degree) for the same family; Section 10 lists polynomial-calculus lower bounds as open.
- Riis 1993, as Theorem 8 of Razborov's survey: degree-d PC refutations when C(m,d) and C(n,d) differ mod p, which gives the matching upper bound p for k = pc, p not dividing c (via Lucas).
- Razborov, Proof complexity of pigeonhole principles (2001), Question 7: the fixed-p, fixed-l bounded-degree question for onto-FPHP^{n+p^l}_n.
- Miksa-Nordstrom (JACM 2024): general PC degree method; onto-FPHP with m = n+1 easy (Riis); the case of characteristic dividing the surplus not covered.

Our prior-art search (2026-11-05) covered:

- Riis, Sitharam, Non-constant degree lower bounds imply linear degree lower bounds, ECCC TR97-048 (1997), pp. 36-38 and 45-46 (Theorem 33, Propositions 27 and 28), opened 2026-10-29
- Beame, Riis, More on the relative strength of counting principles, DIMACS Series 39 (1998), homes.cs.washington.edu/~beame/papers/riis.pdf: Theorem 8.1, Lemma 8.7, Section 10, opened 2026-10-27 and 2026-11-01
- Razborov, Proof complexity of pigeonhole principles, DLT 2001 (LNCS 2295), people.cs.uchicago.edu/~razborov/files/php_survey.pdf: Theorem 8, Question 7, footnote 5, opened 2026-10-27 and 2026-10-29
- Riis, BRICS Report RS-99-29, opened 2026-10-29
- Miksa, Nordstrom, A generalized method for proving polynomial calculus degree lower bounds (CCC 2015; JACM 2024; arXiv 1505.01358), opened 2026-10-27
- de Rezende et al., Theory of Computing 21(6) (2025), opened 2026-10-29
- Lauria et al., Theory of Computing 21(4) (2025), opened 2026-10-27

The checkers' full answers, with every query and file they searched, are published with this finding (`prior-art/`).

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the verification steps below build the published programme library.

How it was produced

Promoted declaring sources are included under `checks/library/`.

How to verify

Lean toolchain: `leanprover/lean4:v4.35.0`. `https://github.com/kodakodapa/autoresearch-findings/tree/2abad0979eeb03ede3a77490f45a4f381119ff02/programmes/proof-complexity`. Install elan, then pin the library subtree while keeping the finding checks:

```
git clone https://github.com/kodakodapa/autoresearch-findings.git
cd autoresearch-findings
git checkout 2abad0979eeb03ede3a77490f45a4f381119ff02 -- programmes/proof-complexity
cd programmes/proof-complexity && lake build
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
AR_LIBRARY=$PWD python -I ../../claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/
```

`checks/run-copies/`: the author's working copies from the run; they import the run's private `Work` package and are not built standalone.

2 Claims

Claim K1

For every prime p, SquareLiftAssumption(p) holds, i.e. there exists an integral square lift on $2p-1$ rows and columns.

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.squareLiftAssumption — pass (`results/K1-1.json`)

Prior art. Partial prior art.

Claim K2

For every prime p and natural c, some T satisfies $\mathrm{Bal}(p, 2p-1, cp^2, T, T, e)$ for every $e \le p-1$, and $p \mid T([]) - 1$.

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_unit_design_uncor — pass (`results/K2-1.json`)

Prior art. Partial prior art.

Claim K3

For every prime p, natural c, and $n \ge 2p-1$, ontoFPHP(n, cp^2) has no degree-p PPCRefutation modulo p (functional encoding, PC line degree).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_no_degree_p_unc — pass (`results/K3-1.json`)

Prior art. Partial prior art.

Claim K4

For every prime p, natural c, and $n \geq 2p-1$, ontoFPHP(n, cp^2) has no standard polynomial-calculus refutation modulo p of line degree at most p (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K1-Unconditional.lean`: Work.UniformC3.uniform_no_standard_degr — pass (`results/K4-1.json`)

Prior art. Partial prior art.

Claim K5

For all natural c and $n \geq 3$, ontoFPHP$(n, 4c)$ has no standard polynomial-calculus refutation modulo 2 of line degree at most 2 (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K5-Main.lean`: Work.UniformC3.no_standard_degree_two — pass (`results/K5-1.json`)

Prior art. Partial prior art.

Claim K6

For all natural c and $n \geq 5$, ontoFPHP$(n, 9c)$ has no standard polynomial-calculus refutation modulo 3 of line degree at most 3 (functional encoding).

Status: proved (axioms: propext, Classical.choice, Quot.sound) ů independently re-proved (skeptic/codex).

Verified. No check scripts.

Evidence.

- proof `checks/run-copies/K5-Main.lean`: Work.UniformC3.no_standard_degree_three — pass (`results/K6-1.json`)

Prior art. Partial prior art.

Provenance. Run `20261102-1249`, claims K1, K2, K3, K4, K5, K6, run-folder commit `359fa20477b8d26a5d2ef8a16e5465474e74b452`, published 2026-11-06.

3 Discussion

This is a lower bound, not an exact value. The best upper bound we know at surplus p^2 is p^2, from Riis's criterion [3, 2]. So for surplus p^2 the degree lies between $p+1$ and p^2. At $p = 2$ the separate exact result that the degree is 4 for surplus 4 ($n \geq 5$) is in the same library. The bound does not grow with ℓ and does not answer Question 7. We have not checked the proof of Riis and Sitharam's Theorem 33, which we cite for comparison only.

4 How to verify

For programme findings, follow the clone, build and run steps in each finding. Each finding has its check scripts in `claims/pc-degree-above-p-onto-fphp-surplus-p-squared/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. Promoted declaring sources are included under `checks/library/`; publication checks import their modules and inspect theorem axioms. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

5 Revisions

- 2026-11-06 — Math rendering on the site; paper lists only cited references
- 2026-11-06 — Library published buildable; verification steps; run copies labelled
- 2026-11-06 — Verification steps; stale run copies removed; encoding stated

References

[1] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[2] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[3] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[4] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.